



Ahsanullah University of Science & Technology

Department of Computer Science & Engineering

Course No : CSE 2214
Course Title : Assembly Language Programming Sessional
Assignment No : 04

Date of Submission : 10/09/2026

Submitted To : Ms. Sumaiya Nuha Mustafina & Ms. Tasnuva
Binte Rahman

Submitted By-

Group : A1
Name : Sakif Ahmed Rafi
Id : 00724105101024
Section : A

Question Selection:

There are 10 questions. Use the following formula to determine the question numbers:

$$Q = (d + i) \bmod 10 + 1$$

Where $i = 1, 3, 5, 7$

d = Last Digit of ID

Calculation:

$d = 4$

For $i=1$, $Q = (4+1) \bmod 10 + 1 = 5 \bmod 10 + 1 = 5+1 = 6^{\text{th}}$

For $i=3$, $Q = (4+3) \bmod 10 + 1 = 7 \bmod 10 + 1 = 7+1 = 8^{\text{th}}$

For $i=5$, $Q = (4+5) \bmod 10 + 1 = 9 \bmod 10 + 1 = 9+1 = 10^{\text{th}}$

For $i=7$, $Q = (4+7) \bmod 10 + 1 = 11 \bmod 10 + 1 = 1+1 = 2^{\text{nd}}$

Question 6: $(4! + 3!) - 2!$ **Solution:**

```
MOV BX,0
```

```
MOV AX,4
```

```
MOV CX,3
```

```
L1:
```

```
    MUL CX
```

```
    LOOP L1
```

```
    ADD BX,AX
```

```
MOV AX,3
```

```
MOV CX,2
```

```
L2:
```

```
    MUL CX
```

```
    LOOP L2
```

```
    ADD BX,AX
```

```
MOV AX,2
```

```
MOV CX,1
```

```
L3:
```

```
    MUL CX
```

```
    LOOP L3
```

```
    SUB BX,AX
```

```
MOV AX,BX
```

```
INT 3
```

Question 8: Find out the average of the ten numbers.

Solution:

```
MOV AX,0
MOV CX,10
L1:
    ADD AX,CX
LOOP L1

MOV BX,10
MOV DX,0
DIV BX
INT 3
```

Question 10: Floor size 80*80, Tiles size 4*4. How many tiles will be required to pave up the floor?

Solution:

```
MOV AX,80
MOV BX,80
MUL BX
MOV CX,AX    ;CX = 80*80

MOV AX,4
MOV BX,4
MUL BX
MOV BX,AX    ;BX = 4*4

MOV AX,CX
MOV DX,0
DIV BX
INT 3
```

Question 2: Temperature conversion from °F to °C (110°F)

Solution:

```
MOV AX,110
SUB AX,32

MOV BX,5
MUL BX

MOV BX,9
DIV BX
INT 3
```